Dynamic accessibility reveals structure in oxygen-relative chemical space

Author: Jaime Ojeda
AI-assisted modeling and analysis using ChatGPT and Grok
E Mail: Jaime.ojse@gmail.com
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1. ABSTRACT

We investigate whether dynamic accessibility in a minimal recoverability-limited system may reveal structure across the periodic table when elements are parameterized relative to oxygen-centered geometry.

Using a simple dynamical model with Gaussian forcing and recovery scaling, we define operational transitions between weak perturbation, damage accumulation, and non-return regimes. A pre-boundary damage mechanism is introduced, allowing systems to accumulate loss of recoverability prior to collapse.

Across 118 elements, parameterized by oxygen-relative radii and chemical-space distance, we observe that elements cluster into distinct dynamical regimes. In particular, a regime where damage precedes non-return appears to emerge robustly across parameter variations.

The relationship between oxygen-relative distance and dynamic accessibility is not well described by a simple linear correlation ($r \approx -0.16$), yet structured clustering persists. Randomizing the geometric embedding removes this structure, suggesting that the observed organization is not an artifact of sampling.

These observations are consistent with the distinction between structural admissibility and dynamic accessibility introduced in a minimal recoverability framework, suggesting that loss of recoverability may precede structural collapse and organize elements into distinct dynamical regimes even when static geometric distance is weak.

All results are derived from a reproducible computational pipeline and are directly linked to publicly available code and datasets.

2. INTRODUCTION

Oxygen occupies a distinctive position in chemical and biological systems, participating simultaneously in bonding, redox exchange, and energy transfer. Recent work has suggested that when elements are embedded in a multivariable chemical space, oxygen defines a non-random local geometry enriched in sulfur-family elements, halogens, and catalytically active metals.

However, such geometric representations remain static and do not directly address whether dynamic behavior across elements reflects similar structure.

In parallel, minimal dynamical models have highlighted a distinction between structural admissibility (what states exist) and dynamic accessibility (which states can be reached under finite-time forcing). In such systems, loss of recoverability may occur prior to structural collapse.

Here, we combine these perspectives by asking whether a minimal recoverability-limited model, parameterized using oxygen-relative geometry, may reveal structured dynamical regimes across the periodic table.

3. MODEL

We extend a previously introduced minimal recoverability model, in which system dynamics are governed by a relaxation term and transient Gaussian forcing:

$$dv/dt = (1 - v)/\tau_r - E(t)$$

where E(t) is a Gaussian pulse parameterized by amplitude (A) and duration (σ).

In this work, each element is embedded into a 2D radial dynamical system derived from this framework. Element-specific parameters are introduced through oxygen-relative geometric scaling:

- r_core defines the intrinsic recovery scale
- r_boundary defines the collapse threshold

Element-specific parameters (r_core, r_boundary) are directly obtained from the input datasets used in the accompanying computational notebook.

The system evolves under the combined effect of recovery and external forcing. Recovery strength decreases as radial displacement increases relative to r_core, allowing each element to express a different accessibility profile under the same forcing sweep.

We introduce a pre-boundary damage mechanism, allowing accumulation of loss of recoverability before the system reaches the collapse boundary. This enables separation between two operational transitions:

- first_damage: onset of loss of recoverability before collapse
- first_nonreturn: crossing of the collapse boundary

Dynamical regimes are defined based on the ordering of these transitions, distinguishing between damage-before-nonreturn, no-transition, and nonreturn-before-damage behaviors.

The implementation of this model, including all parameter sweeps across 118 elements, is provided in the accompanying computational notebook. The notebook uses relative repository paths and requires no access to private external resources.

4. RESULTS

We evaluated the dynamical behavior of 118 elements under a minimal recoverability-limited model parameterized by oxygen-relative geometry.

Each element was assigned a set of parameters (r_core, r_boundary) derived from its position in oxygen-centered chemical space. For each element, we simulated system response across a range of forcing amplitudes and durations, and classified trajectories based on the ordering of two transitions: first_damage and first_nonreturn.

Across the full set of elements, dynamical behavior does not collapse into a single pattern. Instead, elements partition into distinct regimes characterized by different recoverability pathways.

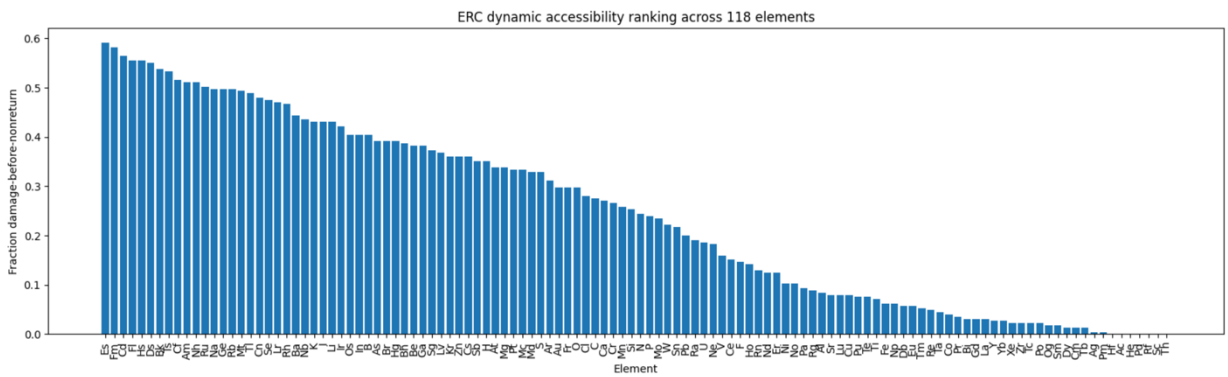

Figure 1— *Global ranking of dynamic accessibility across 118 elements.*
Elements are ranked by the fraction of trajectories exhibiting damage-before-nonreturn behavior under the recoverability model. The distribution reveals a continuous spectrum of dynamic accessibility rather than discrete clustering, with substantial variability across the periodic table.

Figure 1 shows the global ranking of elements based on the fraction of trajectories exhibiting damage-before-nonreturn behavior. Rather than forming discrete clusters, elements span a continuous spectrum of dynamic accessibility, with significant variation across the periodic table.

In particular, a regime in which damage precedes non-return appears across a substantial subset of elements. Other elements exhibit either no transition within the tested parameter range or a regime where non-return occurs before significant damage accumulation.

Importantly, these regimes are not well explained by a simple linear relationship between oxygen-relative distance and dynamic accessibility. While correlation between distance and damage-before-nonreturn fraction is weak ($r \approx -0.16$), elements nevertheless organize into structured patterns when visualized in this space.

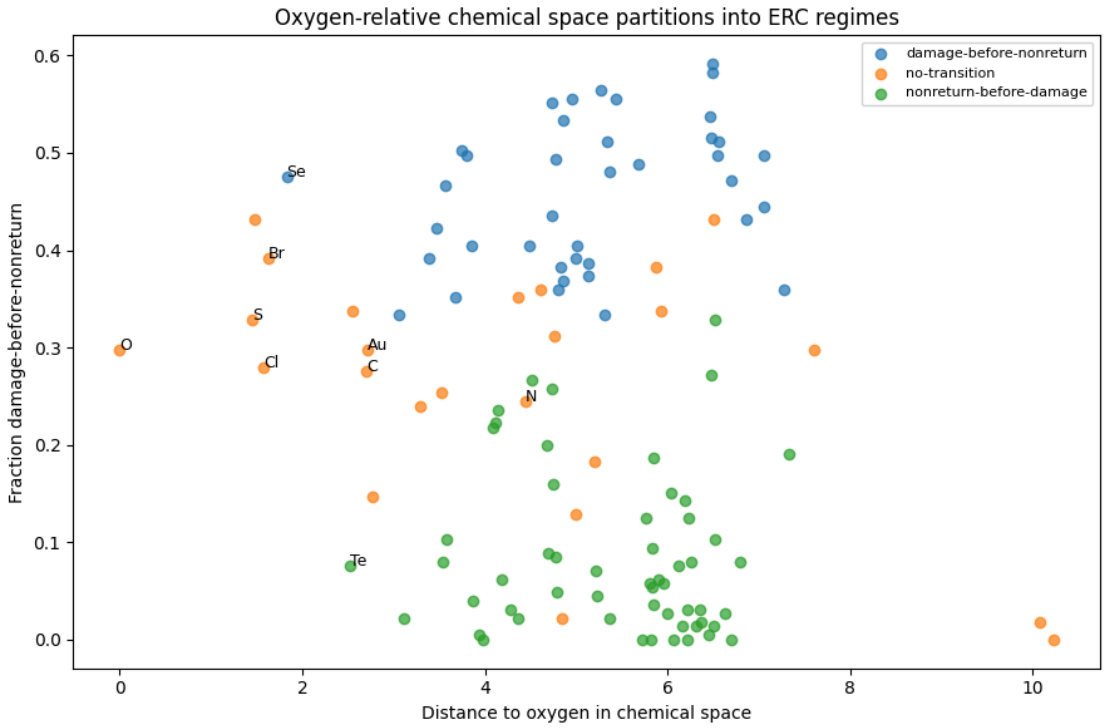

Figure 2 *— Oxygen-relative geometry vs dynamic accessibility reveals structured regimes.*
Each point represents an element positioned by its distance to oxygen in chemical space and its fraction of trajectories exhibiting damage-before-nonreturn behavior. Despite weak linear correlation, elements cluster into distinct dynamical regimes, indicating that accessibility is not determined by geometry alone but constrained by it.

This separation into regimes is not imposed, but emerges from the interaction between geometry and recoverability constraints.

As shown in Figure 2, plotting dynamic accessibility against oxygen-relative distance reveals clear clustering into distinct behavioral regimes. This organization suggests that geometry alone does not determine accessibility, but constrains the dynamical pathways available to each element.

This structure is further supported by a control in which oxygen-relative distances are randomly permuted across elements. Under this perturbation, the clustering observed in the original embedding disappears (Figure 3), indicating that the organization is not an artifact of sampling or statistical noise.

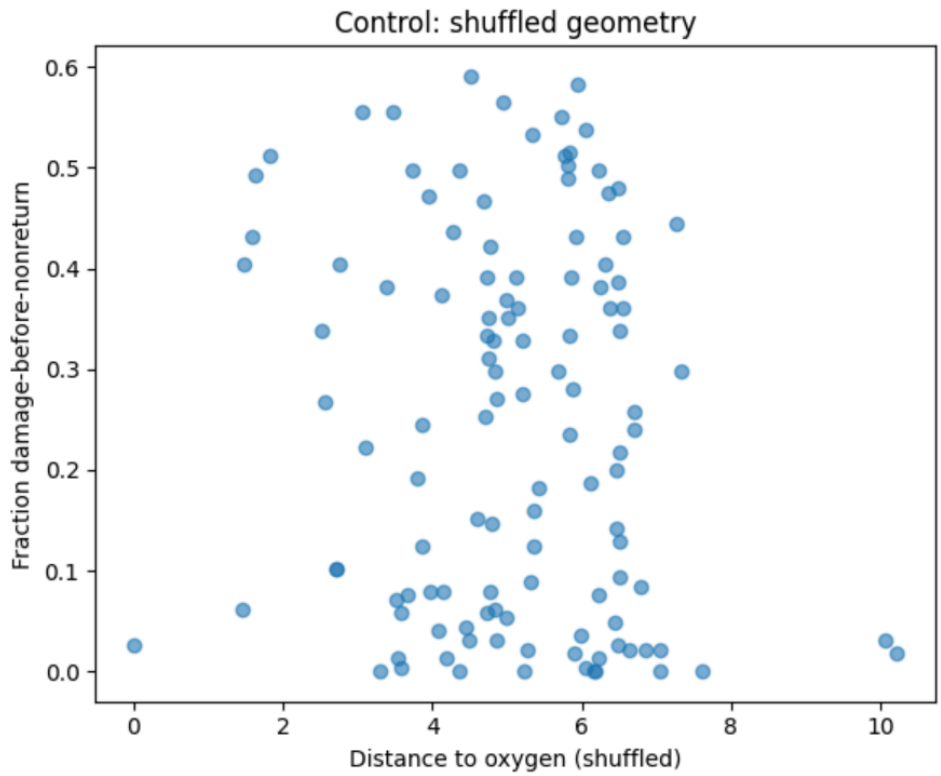

Figure 3— *Control: randomized geometry disrupts dynamical structure.*
Oxygen-relative distances were randomly permuted across elements while preserving dynamic outputs. The structured clustering observed in Figure 2 disappears, indicating that the original organization is not a statistical artifact but reflects a genuine coupling between geometry and dynamical accessibility.

Together, these results suggest that oxygen-relative chemical space may encode distinct dynamical accessibility regimes, not directly reducible to static geometric distance alone.

5. INTERPRETATION

The results suggest that dynamic accessibility is not solely determined by geometric proximity in chemical space.

Instead, the ordering of transitions (first_damage vs first_nonreturn) appears to define regimes that organize elements in a structured way. This organization is not captured by linear correlation, but emerges through pathway-based dynamical behavior.

Structure defines what is possible. Dynamics—through recoverability constraints—defines what is reachable.

Within this framework, oxygen-relative geometry does not uniquely determine dynamical behavior, but appears to constrain the regime in which system dynamics unfold.

Notably, elements with similar oxygen-relative distances can exhibit markedly different accessibility profiles, while elements separated in geometric space may converge to similar dynamical regimes. This decoupling between geometry and accessibility is consistent with the emergence of regime structure observed in Figure 2.

The disappearance of this structure under randomized geometry (Figure 3) further supports the interpretation that the observed organization reflects a genuine coupling between geometry and dynamical constraints, rather than a statistical artifact.

These observations are consistent with the idea that recoverability pathways may reveal latent organization not visible in static representations of chemical space.

6. REPRODUCIBILITY

All simulations, parameter sweeps, and visualizations presented in this work are implemented in a publicly available computational notebook.

The repository includes:

- Input datasets defining oxygen-relative geometry and transition pathway parameters
- A full simulation pipeline applied across 118 elements
- Output tables and figures corresponding to the results presented in this manuscript

The notebook is designed to run using relative repository paths and does not require access to private external resources. All required input files are included within the repository structure.

The computational workflow reproduces:

- Element-wise regime classification based on first_damage and first_nonreturn transitions
- Global ranking of dynamic accessibility (Figure 1)
- Structured clustering in oxygen-relative space (Figure 2)
- Control results under randomized geometry (Figure 3)

Independent replication is supported by the availability of code, data, and documented execution steps.

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